



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ADAPTIVE SOFTWARE ENGINEERING (24CI3201)

A.Y. 2026 - 2027

TITLE OF THE PROJECT	ChangeWise: AI-Powered Adaptive Software Change Impact Analysis	
S. No.	University ID	Name
1	2420030154	G. Anusha
2	2420030288	G. Harini
3	2420030533	P. Chaitra Sree
4	2420030158	M. Pranith Reddy
Name of the Guide		Kalidindi Bhavya Varma

Abstract

1. Problem

Software projects often require changes after they have already been developed and delivered. When a client requests a new feature or modification, developers need to understand which existing modules, dependencies, APIs, databases, and tests may be affected. Identifying these impacts manually can be time-consuming and may lead to missed dependencies, unexpected failures, and inefficient implementation.

2. Proposed Solution

ChangeWise is an AI-powered adaptive software change impact analysis system that connects to an existing GitHub project and analyses its structure, dependencies, and relationships. Developers can enter a new client change request, and ChangeWise evaluates how the requested change may affect the existing software before implementation begins.

3. Key Features

ChangeWise identifies potentially affected components, assigns risk levels based on the extent of the impact, explains the relationships between affected components, and recommends an appropriate implementation order. It also suggests relevant test cases and generates an AI-powered change impact report to help developers plan the requested modification.

4. Benefits

ChangeWise helps development teams make better decisions before modifying an existing system. It reduces the chances of unexpected failures, saves time in impact analysis, improves testing and implementation planning, and supports safer and more adaptive software evolution in response to changing client requirements.

Signature of the Guide

HOD